

Think-Speak-Think

A Process Paper compiled by Rabeeha Adnan and Diamond Nguyen

Thank you Prof. Dr. Kevin Sweet for your consistent insight, time and resources in helping us make Think-Speak-Think real.

Thank you to the staff at Hyphen Space for generously offering us your space.

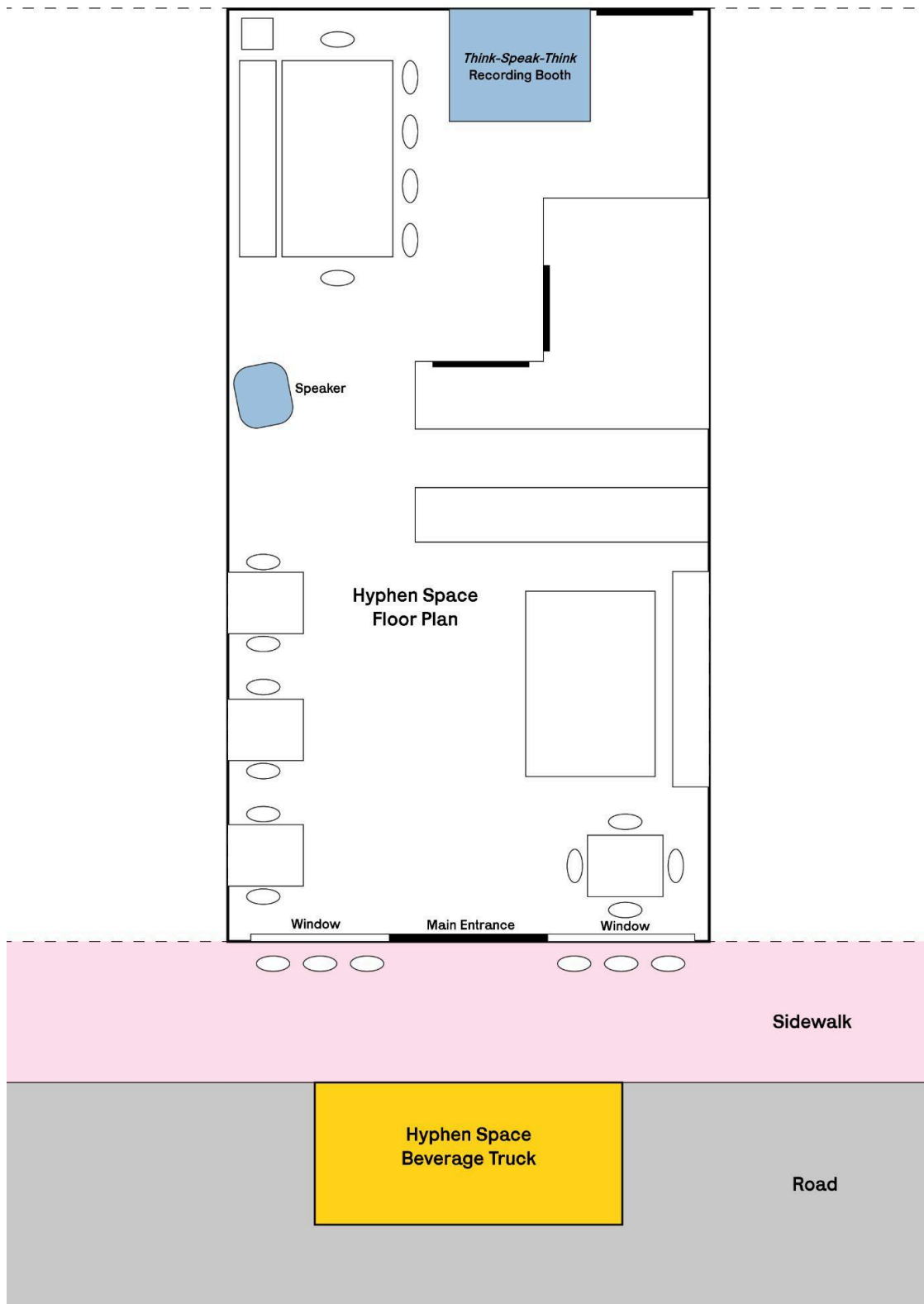
Concept

How do we show up when it seems safer to not?

In an atmosphere of perpetual precarity, surveillance, and dominant narratives that affect how we navigate through the world, showing up becomes deeply political. *Think-Speak-Think*, a collaborative effort by Rabeeha Adnan, Diamond Nguyen, and Neal Ma, responds to this condition of being and invites people to gather together and indulge, frankly. In offering a state of disembodiment, it welcomes people to share, listen, and be.

Think-Speak-Think comprises three essential features: 1) a vocal booth to record one's voice, 2) speakers that project the voice asynchronously in a section of the space which is public, and 3) anonymity.

Our first iteration of *Think-Speak-Think* materialized at Hyphen Space (a creative space and cafe located in Downtown Dallas, TX). We created a small booth that housed our audio recording setup that people could enter individually to record 15 seconds of audio. This setup was connected to a speaker that projected sound to the main area of Hyphen Space (see Floor Plan on the following page).



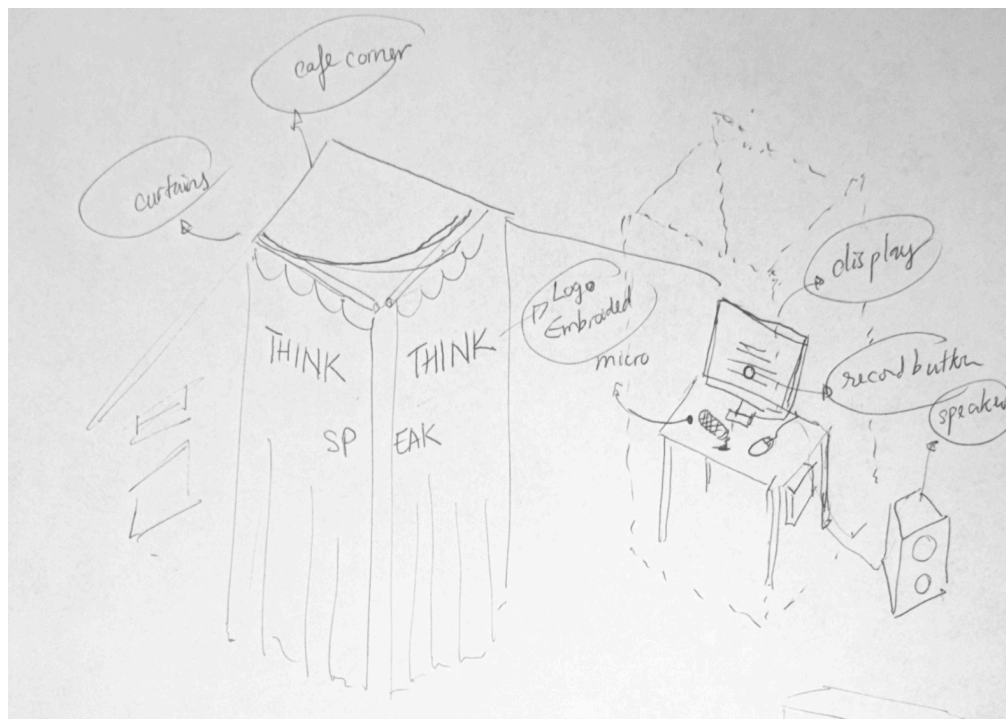
Install Process

We narrowed down two places in Dallas, TX:

1. The Wild Detectives (314 W Eighth St, Dallas, TX 75208), and
2. Hyphen Space (4100 Commerce St #6, Dallas, TX 75226).

Our intention was to create a socially engaged artwork within a public space. The Wild Detectives is an independent bookstore, bar & cafe offering a variety of titles, light fare & drinks, author readings & music. Hyphen Space is a matcha shop with co-working space and a small gallery space. Both the spaces were receptive to us reaching out; however, our timeline and logistics aligned with Hyphen Space and soon after, we had planned the dates for the event.

Since the project had not been produced before, a lot of our initial conversations relied on clear and concise communication about the project and its set up. Hyphen's team was incredibly supportive and believed in our vision despite not having a fully developed or tangible final product.



Initial sketch of the piece

With one day to install, we wanted to ensure that our setup was efficient and stable. While we initially considered repurposing a phone booth, we quickly dismissed that option due to budgetary concerns. With further communication and coordination with the team at Hyphen Space, we resolved the structural logistics and worked around challenges such as the unavailability of corners and limited space.



We utilized a U-shaped curtain rod and two pairs of curtains to create a booth that could comfortably hold one person. To proceed with the curtained booth, we confirmed details such as screwing fixtures and placement with Hyphen Space, and received approval.

Materials list for *Think-Speak-Think*:

- U-shaped rod
- Two pairs of curtains
- Table
- Monitor
- A small 7-inch HD display
- Two pairs of HDMI cables

- Speaker
- Mouse
- Keyboard
- Microphone with stand

In terms of the timeline, we had planned to host the event during the weekend to account for a greater number of people and upon the suggestion of Hyphen's team, we extended the event across three days from April 24 to April 26, 2026 (Friday to Sunday). We had the day prior for installation and the day after for deinstallation. This tight schedule meant that we needed to have every aspect of logistics and resources sorted out in advance, which we did.







Technical Development

The digital interface of *Think-Speak-Think* is built on a lightweight client-server architecture running entirely on a local machine, with no internet dependency during operation by Diamond Nguyen.

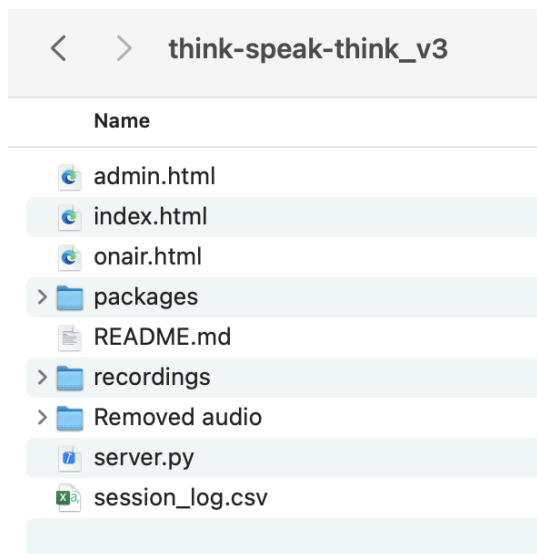
The installation software was developed using Claude (Anthropic) as a coding collaborator. Claude assisted in developing the architecture and writing the full-stack system across Python and JavaScript, including real-time audio recording, pitch processing, normalization, shuffled playback routing, and the browser-based participant interface. Development followed an iterative process in which Diamond directed creative and technical decisions while Claude implemented and revised the codebase accordingly.

Backend (Python): The server is written in Python and handles all audio logic. On launch, it detects available audio devices and routes recording and playback to separate physical outputs: a microphone inside the booth and a speaker in the room. When a participant initiates a recording, the server captures audio through a sound device at 44,100 Hz and applies two processing steps before saving: a pitch shift of -2 semitones via linear interpolation resampling, which lowers the voice slightly to soften

recognizability, and RMS normalization targeting -18 dBFS, which brings all voices to a consistent listening volume regardless of how quietly or loudly someone spoke. Processed recordings are saved as WAV files to a local directory with timestamp filenames, and metadata is logged to a CSV file. A separate thread manages playback, cycling through the recordings folder in a shuffled order with a 3-second gap between each file, refreshing the pool dynamically as new recordings are added or old ones are deleted.

Frontend (HTML, CSS, JavaScript): Three browser-based interfaces are served by the Python HTTP server. The booth UI (index.html) guides participants through a four-state flow: invitation, ready, recording, and thank you, using a 15-second countdown timer and direct API calls to the backend to trigger recording start and stop. The admin panel (admin.html) provides real-time monitoring of recording count, playback status, and pool management controls, including an instant refresh that stops the current file mid-stream. The On Air display (onair.html) polls the server every 1.5 seconds and toggles between an AVAILABLE and IN USE state based on mouse activity detected on the booth screen, providing a visual signal outside the curtained enclosure.

The structure of the project folder is as follows:



Diamond remained in charge of developing the software in conversation with other teammates. In the first version, the prototype only recorded voice and did not apply any modifications. In the second version, the prototype allowed users to listen to and review their recordings before sending them to the speaker, and it also allowed them to manually modify their voice (e.g., change pitch, add reverb). However, after discussions with the team, we decided to discard the review and manual options to make the

process more concise and simple. We then arrived at the third version, in which users record their voices, and the system automatically applies pitch modulation and outputs the recording directly to the sample pool without any review. As a collective, we also agreed to reduce the maximum recording duration from 30 seconds to 15 seconds to make the interaction more direct and efficient.

Visual Language

The visual language of the project was developed by Rabeeha Adnan as we progressed to the outreach stage of the process. We decided on a simple visual style with a minimal color palette consisting of baby blue, deep red, black, and white design elements. Readability was kept clear and the level of information revealed in the graphic assets was intentionally minimal.

The visual language was then adapted for social media, tabloid posters, the assets for the booth, and the recording interface. Rabeeha remained in charge of readapting and reproducing graphic assets for community outreach through social media and physical posters.

All posters and interface components are shared in the next few pages:

Community Activation • hosted
by Hyphen Space

Think
Speak
Think

a social project by
Rabeeha Adnan Diamond Nguyen Neal Ma

Friday, April 24, 2026 — Sunday, April 26, 2026
4 — 7 pm
4100 Commerce St #6, Dallas, TX 75226

How do we show up when it seems safer to not? In an atmosphere of perpetual precarity, surveillance and dominant narratives that affect how we navigate through the world, showing up becomes deeply political.

Taina
Speak
Taina

Indulge, frankly.

Speak your mind, read a poem,
share a prayer, bend the rules, question your
mother or the government, prompt or provoke,
confess and confront.

a social project by
Rabeeha Adnan
Diamond Nguyen
Neal Ma

Think Speak Think

You may record your voice privately inside the booth.

Once recorded, your voice will be modulated and will enter the series of voices in the room in random order.

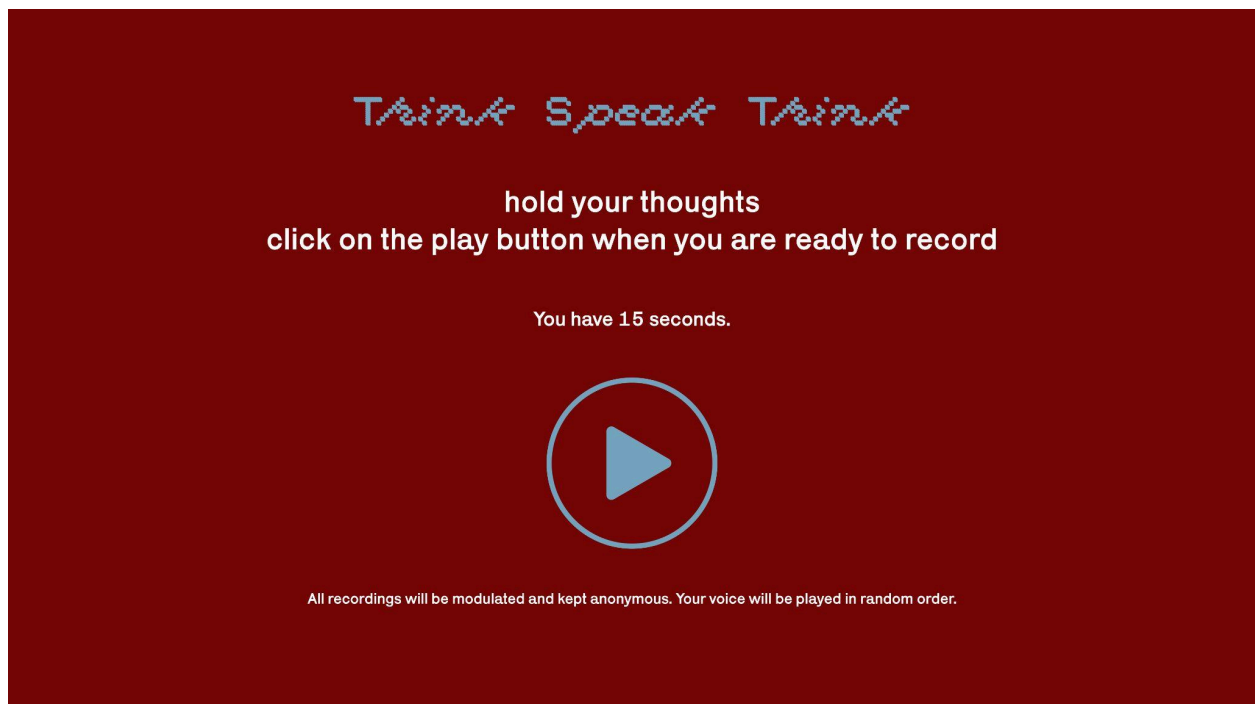
Your voice will not correspond to your turn in the booth. This will ensure your anonymity.

The series of voices will continue playing in the room.

We invite you to be authentic, playful and creative.

INSTRUCTIONS

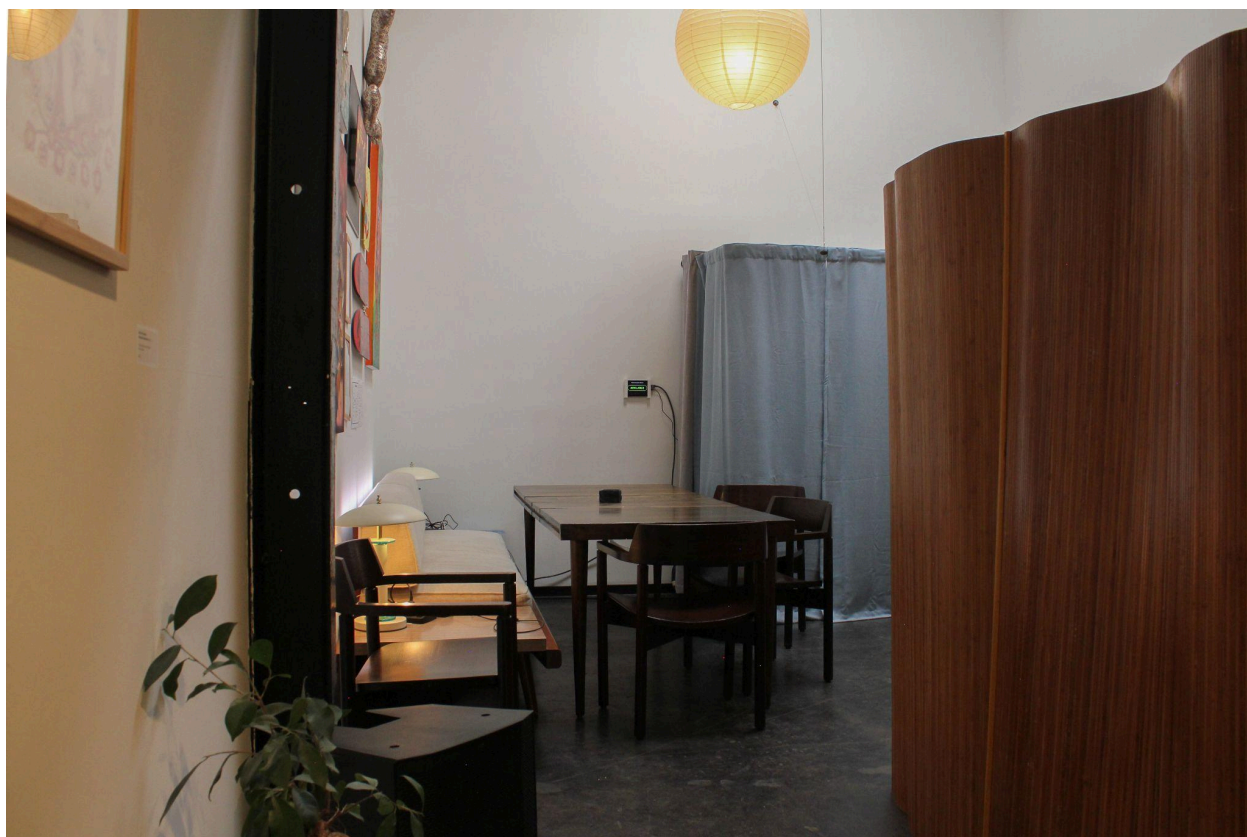
1. Ensure the booth is empty before you enter using the sign located outside the booth which will show whether it is in use or available.
2. Once you enter, think of what it is you want to say.
3. Use the mouse to navigate the screen.
4. Press "Enter the Room" and the screen will redirect you to the recording stage.
5. Press Play button to start recording.
6. You will have 15 seconds to record your voice.
7. You may stop recording at any point.
8. You may step out when you are done and listen. Your voice is now part of the room.
9. Thank you for participating.

Interface:



Documentation from the space found below:









Reflections and Challenges

Overall, both the proposal and installation stages proceeded smoothly in a step-by-step manner. We were able to resolve most of the issues that arose—software and hardware both. We were able to successfully incorporate the essential component of anonymization into the system and achieved the expected results.

The staff at the space was highly supportive throughout the entire process, from the proposal stage to execution and deinstallation. This allowed the collaboration to flow smoothly. However, since the space embodies multiple identities, such as a café, coworking space, gallery, and event venue, introducing an interactive booth is not inherently typical in this environment, which led to some challenges.

A space mediates the content of the information of the information people share. The way people perform in a space is mediated by its existing norms, which in turn affect how they approach and engage with the work. As a coworking space, people are often using headphones, working individually, or interacting in small groups. They may not be aware of the ongoing *Think-Speak-Think* interactive booth and may feel hesitant to participate. Most people feel reluctant to try something unfamiliar if they are by themselves versus when they are in groups of friends. Additionally, the difference between times of day, especially day time and night time influences how freely people perform in public spaces. Since the installation remained open from 4 pm to 7 pm, before sunset in Dallas, TX during May, people were more focused on work rather than feeling relaxed and open to engaging with the interactive booth.

The location of the interactive booth within the space affected audience engagement. We initially wanted to place the speaker farther away from the booth— either on the street or in the main seating area, as people tend to line up outside to get their drinks. However, our booth was located at the back of the space and the speaker was neither visible nor audible towards the entrance. In addition to procuring resources that would allow us to extend the speaker farther towards the entrance, our project would have benefitted from repositioning of the audio source, such that it orientates people towards itself.

Adjusting the audio volume was also a challenging aspect. We needed to ensure that the sound was not overshadowed by the overall ambient noise, while also avoiding

disturbing people who were working, so that those who were socializing could still comfortably engage with the installation.

The public at Hyphen Space were not aware of the experience at first until we informed and encouraged them to try it. This is primarily due to the fact that the café counter is located outside the premises; this meant that we lost 70 percent of the public that only ordered drinks outside and left. Hence, a large portion of the public remained unaware of the event despite posters, unless it was communicated verbally. This aspect of outreach can and will be revisited in future iterations, with relocation and/or reconfiguration of the audio source, more visible signage and in-person guidance to enhance awareness. The graphic design campaign could be better aligned with how people enter and move through the space. Additionally, the majority of Hyphen Space's customers are Asian American, and they tend to be more reserved and hesitant to participate. There is an added level of anxiety associated with approaching novelty, particularly interactive installations such as *Think-Speak-Think*.

Regarding the form of interaction, the setup was easy for viewers to engage with, based on observed interactions and feedback from interviews. We observed that people found it easier to approach the booth when given a prompt instead of having to come up with something on their own, especially since the instructions were more broad than specific.

Many participants chose to speak in their mother tongue, such as Chinese, Vietnamese, Arabic, and others. Within the context of Hyphen Space, most contributions were positive and playful, including people reading poems, sharing an anecdote, an immediate thought or feeling. There was one instance that was flagged as inappropriate after a day by the staff. We removed those audio clips from the playback loop but kept them in the archive, which pointed to different markers to indicate what is considered inappropriate.

The installation sparked curiosity and conversations through both interaction and listening to the recordings. For example, after hearing poems read in Vietnamese, one audience member felt encouraged to step in and shared her own poem. People often laughed and enjoyed the more humorous recordings. Overall, excitement towards interaction increased once the initial barrier of unfamiliarity dissipated.

Future considerations

As we reflect on the development and execution process of the first iteration of *Think-Speak-Think*, we successfully managed to fulfil our core idea. We have further been able to assess how we can improve *Thin-Speak-Think* in its future iterations. As the project is modular and mobile, it can reconfigure in various spaces, allowing its tone and content to adapt accordingly.

We observed an element of anticipation and gratification. Participants desire to hear their voices after recording — we can further explore how to build anticipation and deliver that expected sense of gratification. As more voices are recorded, the wait time increases before participants hear their own recordings.

Other insights include speaker placement and the relationship between what people say and the chosen space and time (e.g., a White House setting vs. a coffee shop, bar, or gallery, shaped by the establishment, neighborhood, and time of day). We can improve the audio quality by making changes to (a) the microphone equipment, (b) soundproofing the booth, or (c) audio processing within the Python script, or a combination of all three. We can also consider transducing the sound or using a directional microphone. Additionally, it is important to think about how people encounter the sound and the flow of the interaction.

The booth setup raises questions about relocation. Keeping the setup consistent across locations reveals how different spaces affect the piece, while adapting the setup to each new location may instead reveal qualities of the space itself. Finally, we can incorporate embodied approaches, such as having a person physically present to welcome and invite participation as we noted that verbally inviting people to participate made them less apprehensive and more curious. Engaging with the installation often requires multiple layers of permission, and reducing these barriers could encourage greater involvement.
